Lab Exercise 16- Creating a ConfigMap from a Folder

Name- Manish kumar

SAP ID- 500119723

Batch-2

In this lab exercise, you will learn how to create a ConfigMap in Kubernetes from a folder containing multiple files. This allows you to manage multiple configuration files as a single ConfigMap.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

Step 2: Prepare the Folder with Multiple Files

Create a directory named configmap-folder:

```
mkdir configmap-folder
cd configmap-folder/
```

```
PS D:\kubs-misha> mkdir configmap-folder


    Directory: D:\kubs-misha


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        17-03-2026     14:18                configmap-folder


PS D:\kubs-misha> cd configmap-folder
```

Create multiple configuration files inside the directory:

config1.txt with content:

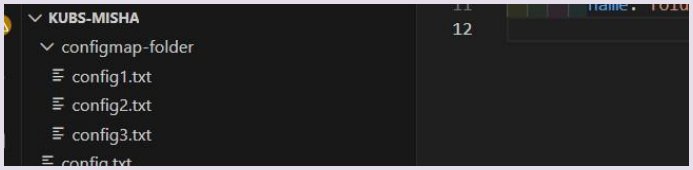
```
key1=value1
```

config2.txt with content:

```
key2=value2
```

config3.txt with content:

```
key3=value3
```



Verify the directory structure:

```
ls configmap-folder/

cd ..
```

```
PS D:\kubs-misha\configmap-folder> cd ..
PS D:\kubs-misha> ls configmap-folder\


    Directory: D:\kubs-misha\configmap-folder


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        17-03-2026     14:19             11 config1.txt
-a----        17-03-2026     14:19             11 config2.txt
-a----        17-03-2026     14:20             11 config3.txt
```

You should see the files config1.txt, config2.txt, and config3.txt.

Step 3: Create a ConfigMap from the Folder

Create a ConfigMap named folder-configmap from the folder:

```
kubectl create configmap folder-configmap --from-file=configmap-folder/
```

```
PS D:\kubs-misha> kubectl create configmap folder-configmap --from-file=configmap-folder/
configmap/folder-configmap created
```

Verify the ConfigMap:

```
kubectl get configmap folder-configmap -o yaml
```

```
PS D:\kubs-misha> kubectl get configmap folder-configmap -o yaml
apiVersion: v1
data:
  config1.txt: key1=value1
  config2.txt: key2=value2
  config3.txt: key3=value3
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-17T08:51:17Z"
  name: folder-configmap
  namespace: default
  resourceVersion: "6010"
  uid: 01e2a8a9-bc4c-47cd-b6ef-807ac12a5c9a
```

You should see the folder-configmap with the contents of all the files in the configmap-folder.

Step 4: Use the ConfigMap in a Pod

Create a file named **pod-folder-configmap.yaml** with the following content:

```
apiVersion: v1 kind:
Pod metadata:
  name: configmap-pod
spec:   containers:   -
name: nginx     image:
nginx        envFrom:     -
configMapRef:
       name: folder-configmap
```

In this manifest:

- The Pod mounts the folder-configmap at /etc/config.
- The busybox container concatenates and prints the contents of the files.

Apply the manifest to create the Pod:

kubectl apply -f pod-folder-configmap.yaml

```
  uid: 01e2a8a9-bc4c-47cd-b0ef-807ac12a5c9a
PS D:\kubs-misha> kubectl apply -f pod-folder-configmap.yaml
pod/configmap-pod created
```

Check the status of the Pod:

kubectl get pods

```
pod/configmap-pod created
PS D:\kubs-misha> kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          9s
PS D:\kubs-misha>
```

View the Pod logs to see the content of the mounted files:

kubectl exec -it configmap-pod – printenv

```
PS D:\kubs-misha> kubectl exec -it configmap-pod -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
NGINX_VERSION=1.29.6
NJS_VERSION=0.9.6
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
config1.txt=key1=value1
config2.txt=key2=value2
config3.txt=key3=value3
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
TERM=xterm
HOME=/root
```

You should see the contents of config1.txt, config2.txt, and config3.txt printed.

Step 5: Clean Up

After completing the exercise, clean up the resources created:

```
kubectl delete pod folder-configmap-pod

kubectl delete configmap folder-configmap

rm -r configmap-folder
```

```
PS D:\kubs-misha> kubectl delete pod folder-configmap-pod
Error from server (NotFound): pods "folder-configmap-pod" not found
PS D:\kubs-misha> kubectl delete configmap folder-configmap
configmap "folder-configmap" deleted from default namespace
PS D:\kubs-misha> rm -r configmap-folder
```

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